

RESEARCH PROFILE

PhD researcher working primarily on open quantum systems and non-equilibrium quantum dynamics. Research interests include quantum many-body dynamics, relaxation and transport, dissipation and decoherence, non-Hermitian physics, statistical mechanics, quantum information, quantum simulation, and connections with machine learning.

EDUCATION AND AFFILIATION

Aug. 2022 - Present	PhD Candidate in Physics Institute of Physics, Chinese Academy of Sciences · Beijing, China
Aug. 2018 - Jun. 2022	B.Sc. in Physics, Honors Program in Science Jilin University · Jilin, China GPA: 3.88/4.00 (93.2/100).

RESEARCH INTERESTS

- Non-equilibrium quantum dynamics
- Quantum many-body physics
- Statistical mechanics
- Open quantum systems
- Non-Hermitian physics
- Quantum information and simulation

PUBLICATIONS AND PREPRINTS

2026	Geometry-adaptive formulation of non-Bloch bands in arbitrary dimensions and spectral instability ↗ Ze-Yu Xing , Yuncheng Xiong, and Haiping Hu <i>Communications Physics</i> 9, 127 (2026) Ze-Yu Xing and Yuncheng Xiong contributed equally.
2025	Universal spreading dynamics in quasiperiodic non-Hermitian systems ↗ Ze-Yu Xing , Shu Chen, and Haiping Hu <i>Physical Review B</i> 111, L180203 (2025)
2024	Non-Hermitian skin effect in arbitrary dimensions: non-Bloch band theory and classification ↗ Yuncheng Xiong, Ze-Yu Xing , and Haiping Hu <i>Preprint, arXiv:2407.01296</i> (2024)

HONORS AND AWARDS

2020	Ny Tsi-Ze Scholarship Institute of Physics, Chinese Academy of Sciences
2020	Honorable Mention, Mathematical Contest in Modeling Consortium for Mathematics and Its Applications
2019	National Scholarship Ministry of Education, People's Republic of China
2019	Outstanding Student of Jilin University Jilin University